Alternative Splicing And Global Transcriptome Changes Associated With LPS Stimulation In Human Peripheral Blood Mononuclear Cells

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Introduction

Lipopolysaccharide (LPS), a key component of gram-negative bacterial cell walls, is a potent activator of the innate immune system and widely used as a model to study inflammatory responses. While the transcriptional response to LPS stimulation has been characterized, the role of alternative splicing (AS) in modulating this response remains largely unexplored. This study aims to comprehensively characterize the transcriptomic effects of LPS by utilizing deep RNA sequencing to evaluate both differential gene expression and alternative splicing associated with LPS treatment. Additionally, it seeks to uncover underlying mechanisms of immune regulation to identify potential therapeutic targets.

Methods

Sample Preparation, RNA Sequencing, and Alignment

Peripheral Blood Mononuclear Cells (PBMCs) from three healthy female donors were treated with 100 ng/mL of LPS or PBS as a control for 24 hours. Following treatment, RNA was isolated, purified, and sequenced using paired-end reads (150 bp) on an Illumina NovaSeqX platform. Sequencing reads were quality-checked with FASTQC and MultiQC, trimmed using Trimmomatic, and aligned to the GRCh38 reference genome using the STAR aligner. Gene abundance and transcript levels were quantified using featureCounts and TPMcalculator. The data are accessible through GEO Series accession number GSE308443

Differential Gene Expression Analysis

Differentially expressed genes (DEGs) were identified using edgeR, applying a strict false discovery rate (FDR < 0.05). Pathway impact analysis was performed using KEGG and STRING databases to identify perturbed pathways. Pathway topology and gene-gene interactions were incorporated to determine perturbed pathways associated with LPS treatment.

Cell Type Composition Estimates

Cell-type deconvolution was performed using TPM estimates through the immunedeconv R tool, employing the quantTIseq method to assess changes in immune cell populations. Two-sided Wilcoxson's tests were used to test statistically significant differences between two groups.

Differential Alternative Splicing Analyses

rMATS-turbo was used to detect alternative splicing events, including skipped exons (SE), retained introns (RI), mutually exclusive exons (MXE), alternative 3' splice sites (A3SS), and alternative 5' splice sites (A5SS). Events were filtered for significance (FDR < 0.01, |ΔPSI| > 0.05). Functional enrichment of spliced genes was performed using NEASE and DIGGER, integrating protein-protein interaction networks and pathway annotations from Reactome, BioPlanet, and WikiPathways. A splice-aware network-based enrichment approach was applied to identify rewired interactions in immune-related pathways. Isoform-specific impacts on protein domains and binding sites were evaluated for enriched pathways.

Results

Differential Gene Expression

A total of 490 DEGs were identified between LPS-treated and control PBMCs. Pathway impact analysis revealed 46 significantly perturbed KEGG pathways, including Toll-like receptor signaling and PI3K/AKT signaling. Two novel pathways, SNARE interactions in vesicular transport and the mRNA surveillance pathway, were identified as key regulators of LPS responses.

Cell Type Composition

LPS treatment was associated with an increase in pro-inflammatory M1 macrophages and a decrease in anti-inflammatory M2 macrophages. No significant changes were observed in other immune cell types, such as T cells or monocytes.

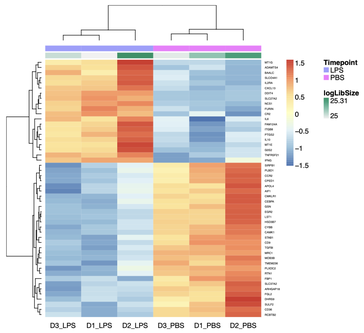

Figure 1. Heatmap of the top 50 differentially expressed genes between LPS and PBS-treated PBMCs from three women donors identified in the global transcriptome analysis. Rows and columns were clustered using the Euclidean distance and the complete linkage method.

Differential Alternative Splicing

A total of 90,364 splice events were identified in the LPS and PBS-treated PBMCs. Of these, 1,316 skipped exon (SE), 388 retained intron (RI), 347 mutually exclusive exon (MXE), 279 alternative 3' splice (A3SS), and 167 alternative 5' splice (A5SS) site events were significant. A total of 1,634 unique genes were identified across the 2,507 significant splice events, with SE being the most frequent splicing type. Among these, 70 genes were both differentially expressed and differentially alternatively spliced. Of the 1,316 SE events, 698 were downregulated and 618 were upregulated in the LPS-treated PBMCs. Additionally, 156 DAS events were identified across 88 long non-coding RNA genes. Two lncRNA genes (MALAT1, PVT1) were identified as DAS in our study that are concordant with previous findings of alternative splicing. Splice-aware network analysis identified 48 WikiPathways and 70 Reactome pathways significantly enriched for AS exons. Enriched pathways were found for 6 categories of biological functions related to Immune System and Inflammation.

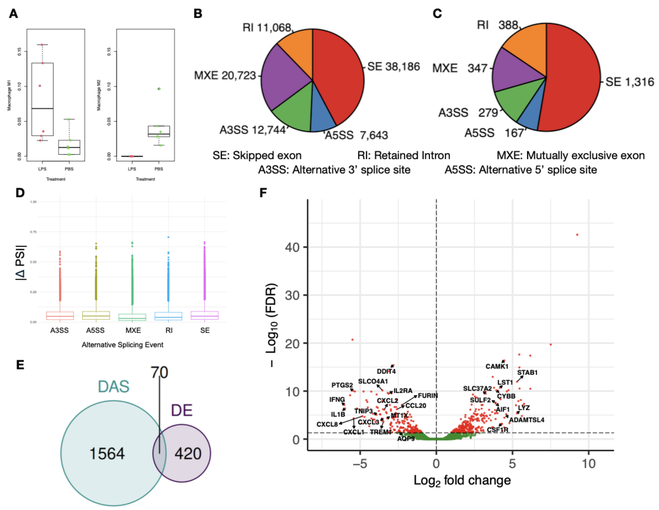

Figure 2. (A) Differences in the abundance of Macrophage M1 and M2 cell types by treatment estimated by immune deconvolution (quanTIseq). (B) Summary pie chart of total alternative splicing events identified in the LPS- and PBS-treated cells after filtering for high read support and extreme PSI values. (C) Summary pie chart of differential alternative splicing events between the LPS and PBS treatments (FDR <0.01, |deltaPsi| ≥0.05). (D) Box plots of the absolute value of deltaPSI by event type. (E) A Venn diagram depicts the overlap of differentially expressed (DE) genes and differentially alternatively spliced (DAS) genes across all splicing events. (F) Volcano plot of DE genes between LPS and PBS-treated donor PBMC samples. Significantly DE genes are colored red (FDR < 0.05), non-significant genes are colored green. Space permitting, a number of genes that are both DE and DAS are labeled for reference.

Table 1. A summary of the annotated lncRNA genes that were identified to have differential alternative splicing in human PBMCs stimulated with LPS as compared to PBS.

DAS Events (n)	Genes (n)	Gene symbols (DAS event)
1	58	C8orf44 (SE), CHROMR (RI), CYTOR (SE), DLEU2 (RI), DPH5-DT (MXE), ENSG00000233461 (SE), ENSG00000254641 (SE), ENSG00000258181 (SE), ENSG00000267787 (SE), ENSG00000273189 (A3), ENSG00000284707 (SE), ENSG00000285756 (MXE), ENSG00000286044 (SE), ENSG00000290816 (A5), ENSG00000290832 (SE), ENSG00000291060 (RI), ENSG00000291067 (MXE), ENSG00000291068 (RI), ENSG00000291100 (SE), ENSG00000291105 (SE), ENSG00000291136 (SE), ENSG00000293154^ (SE), FAM111A-DT (A3), FMNL1-DT (A3), FTX (SE), GARS1-DT (SE), GBA1LP (RI), GMDS-DT (SE), GOLGA2P5 (SE), HISLA (A5), IQCH-AS1 (MXE), JPX (RI), LGALS17A (SE), LINC00662 (A5), LINC00680 (SE), LINC00944 (SE), LINC01237 (SE), LINC01410 (A5), LINC03049 (A5), MAP3K2-DT (RI), MFSD14CP (SE), MUC20-OT1 (SE), NAPSB (MXE), NUTM2A-AS1 (A3), PVT1 (MXE), RELA-DT (RI), RPL23AP7 (A5), SDHAP1 (SE), SIMALR (SE), SNHG12 (RI), SNHG14 (SE), SNHG26 (RI), STAG3L5P-PVRIG2P-PILRB (MXE), TMEM161B-DT (SE), TPT1-AS1 (SE), TRAF3IP2-AS1 (SE), WAKMAR2 (SE), ZNF213-AS1 (SE)
2	16	AFG3L1P (SE), C22orf46P (SE), ENSG00000291055 (RI), ENSG00000291174 (A3, RI), ENSG00000291203 (MXE), GIHCG (SE), HILPDA-AS1 (MXE), LEF1-AS1 (SE), LINC00174 (SE), LINC00987 (RI, SE), LINC01128 (MXE, SE), LINC02328 (SE), MITA1 (SE), NSUN5P2 (RI, SE), PCBP1-AS1 (MXE), SLC30A4-AS1 (MXE, RI)
3	7	ADCY10P1 (A5, SE), CEP250-AS1 (SE), ENSG00000293497 (SE), NOP14-AS1 (A3, RI, SE), POLR2J4 (A5, MXE, SE), PRKCQ-AS1 (MXE, SE), SNHG16 (A5, RI, SE)
4	4	ENSG00000284697 (MXE, SE), MALAT1 (A3, SE), MRPL20-AS1 (SE), PELATON (A3, SE)
7	1	ENSG00000242588 (MXE, RI, SE)
10	1	PSMA3-AS1 (A3, MXE, SE)
12	1	GUSBP11 (MXE, SE)

Abbreviations: ^, global differential expression; A3SS, alternative 3' splice sites; A5SS, alternative 5' splice sites; AS, alternatively spliced; MXE, mutually exclusive exons; RI, retained introns; SE, spliced exon.

Discussion

A total of 490 differentially expressed genes were identified between LPS-treated and control PBMCs. Pathway impact analysis revealed 46 significantly perturbed KEGG pathways, including Toll-like receptor signaling and PI3K/AKT signaling. Two novel pathways, SNARE interactions in vesicular transport and the mRNA surveillance pathway, were identified as key regulators of LPS responses. These findings demonstrate the transcriptional complexity of LPS-induced inflammation and highlight key genes driving the immune response, such as TLR5, IRAK2, and NFKBIA.

The macrophage polarization is consistent with prolonged LPS-induced immune activation and highlights macrophage dynamics as a critical component of the inflammatory response.

We identified 2,507 significant AS events across 1,634 unique genes, highlighting the widespread impact of AS on immune-related pathways. Notably, pathways such as Toll-like receptor signaling and PI3K/AKT network regulation were enriched for AS events, suggesting splicing as a mechanism to fine-tune inflammatory signaling. Novel pathways, including neddylation and focal adhesion, indicate AS may also regulate protein degradation and immune cell migration. Additionally, long non-coding RNAs, such as MALAT1 and PVT1, were found to undergo AS, further emphasizing their regulatory role in inflammation. These findings provide new insights into post-transcriptional gene regulation during immune responses and offer potential therapeutic targets for inflammatory diseases and treatment effects (e.g., fatigue, neuropathy).

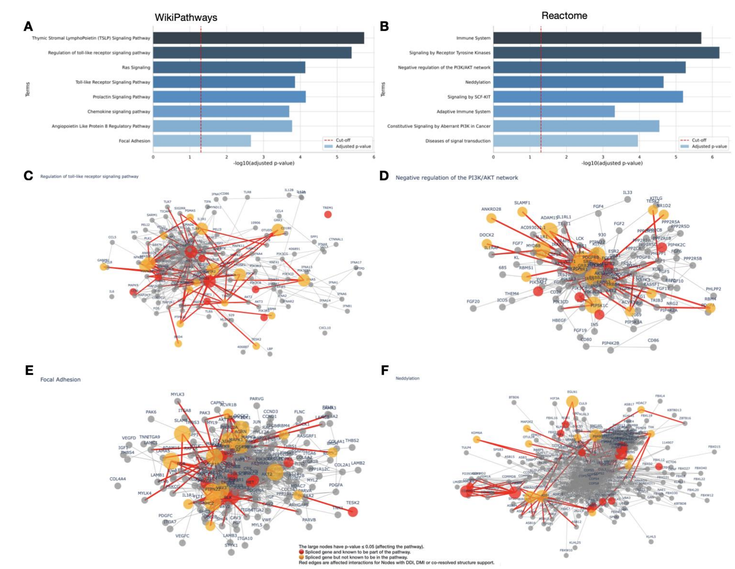

Figure 3. Splice-aware pathway protein-protein interaction (PPI) network enrichment analysis of exon skipping events. Top eight enriched pathways from the (A) WikiPathway and (B) Reactome databases. Visualization of the spliced genes on the PPI network for the pathways: (C) Regulation of toll-like receptor signaling pathway (D) Negative regulation of the PI3K/AKT network, (E) Focal Adhesion, and (F) Neddylation.

Conclusions

This study provides the first transcriptome-wide characterization of alternative splicing in LPS-stimulated PBMCs, highlighting its critical regulatory role in innate immune responses. Alternative splicing influences key pathways, such as Toll-like receptor signaling and macrophage polarization, and rewires protein-protein interactions and regulatory networks, offering novel insights into immune modulation. Additionally, long non-coding RNA splicing and novel pathways, such as SNARE interactions, present opportunities for therapeutic exploration in immune-related diseases and treatments.

Acknowledgements

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